COMPUTERS

Week 2

MACHINES AND COMPUTATIONAL MODELS

In this section, you will look at how the data that is entered into a computer system is processed and the results are output for the user.

OBJECTIVES

- Understand the input-process-output model
- Understand that there is a range of computational models (sequential, parallel and multi-agent)

HOW MANY OF THESE THINGS DO YOU THINK OF AS COMPUTERS?





COMPUTER

is a machine that takes some kind of **input** from its surroundings, **processes** the input according to given rules, and provides some kind of **output**.

ACTIVITY 1

INPUTS AND OUTPUTS

- Name any inputs and outputs for the provided things in slide 4

SUBJECT VOCABULARY

Input: to enter data into a computer

Process: to change the meaning or format of some data

Output: to display or output data that has been processed (or has been stored)

SUBJECT VOCABULARY

Function machine: a metaphor or diagram that represents a machine that takes an input. It applies a rule such as a set of operations and delivers the answer as an output



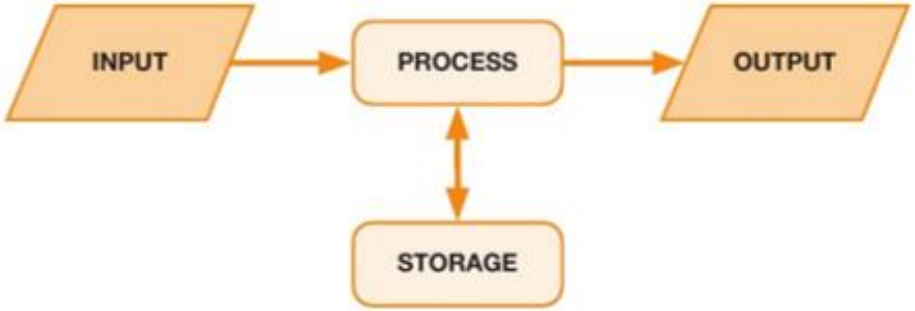
A simple maths function



Indicates an input or output.



Indicates a process to be carried out.



A SIMPLE FUNCTION MACHINE WITH STORAGE

As your algorithms become more complicated, it can be useful to show the steps in a flowchart.

RANGE OF COMPUTATIONAL MODELS

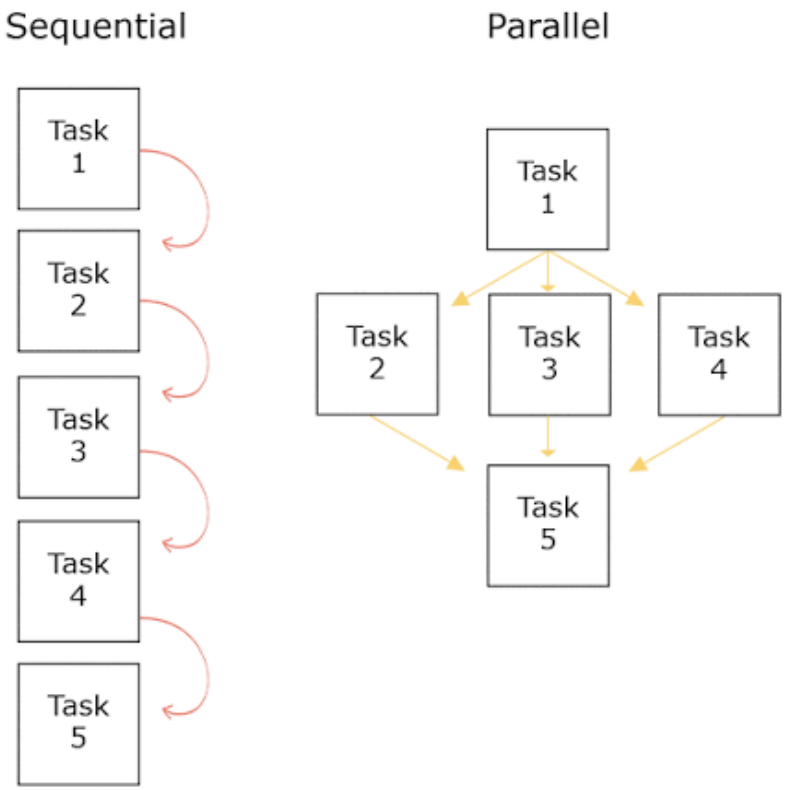
Computational models describe **how** algorithms are executed by a machine, e.g. by processing data using inputs and outputs. There are different models of how this processing can be carried out

SEQUENTIAL

- In the sequential model, this is the way that we commonly imagine how computers work. This involves following instructions in an algorithm step by step, in order, from start to finish.

PARALLEL

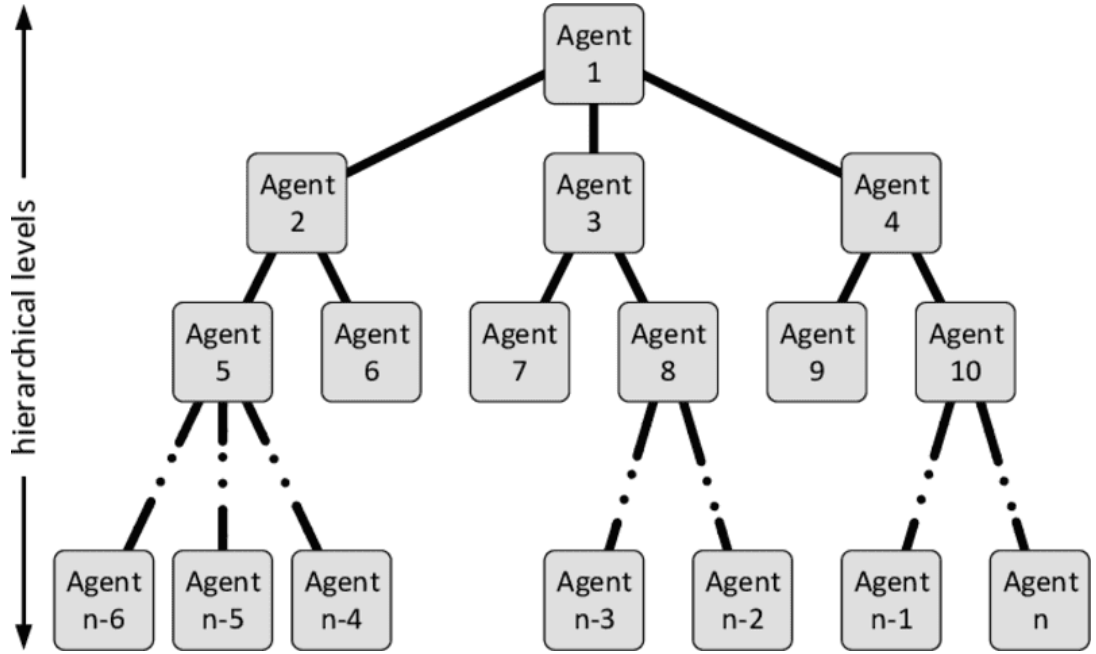
- **Computer processes are distributed between two or more installed processors in a computer.**
- Each separate part of the algorithm that each processor processes can be combined.
- It requires an operating system capable of supporting two or more processors.
- It also requires software programs capable of distributing processes between them equally.


Sequential
Task 1
Task 2
Task 3
Task 4
Task 5
Parallel
Task 1
Task 2
Task 3
Task 4
Task 5

COMPARISON

MULTI-AGENT MODEL

- **Separate tasks or algorithms are processed by different systems (agents) to perform a particular function.**
- Agent a computer system that can interpret its environment through its sensors.
- It can act autonomously upon that environment (through its effectors (something that produces an effect by carrying out an action))
- Each agent acts independently and under its own control.
- Each agent is autonomous (self-directed), but they cooperate with each other through negotiation and coordination.


hierarchical levels
Agent 1
Agent 2
Agent 3
Agent 4
Agent 5
Agent 6
Agent 7
Agent 8
Agent 9
Agent 10
Agent n-6
Agent n-5
Agent n-4
Agent n-3
Agent n-2
Agent n-1
Agent n

EXAMPLE

SUMMARY

- Although computers come in all sorts of shapes and sizes, they all mainly do the same thing. They take in inputs, process them according to a given set of rules, and generate outputs.
- There are different models of different complexity for the processing of data by computer systems. Algorithms expressed in program code specify the processes a computer carries out.

CHECK PAGES (160 - 163)